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Dear Editor,

I am pleased to submit the manuscript “**A Resolution Framework for Finite-Sample Model Selection**” for consideration as a regular paper in the *International Journal of Data Science and Analytics*.

The problem. Model selection across data science follows one default rule: evaluate candidates on a held-out set and keep the best score. This tacitly assumes that 55% is genuinely better than 52%. Whether that assumption holds is a property of the evaluation budget, not of the models—yet it is almost never checked before evaluation resources are committed. Existing tools (paired t -tests, significance guidelines, post-selection inference) are post-hoc: they diagnose the problem only after the experiment is complete.

The contribution. I develop $\Delta_{\min}$, a pre-experimental resolution diagnostic that returns the smallest performance gap a given budget N can distinguish, requiring only N and an approximate accuracy level. Around this threshold the paper builds an operational protocol: a decision rule for which I prove the exact condition under which uniform model averaging dominates argmax selection in expected regret; bootstrap and Monte Carlo calibration linking the threshold to conventional 80% and 95% power; an extension to continuous metrics via the Delta method; a catalog of nine reproducible diagnostic pitfalls; and a one-page quick-reference table.

I am explicit in the manuscript that the underlying expression is classical power analysis. The contribution is its conversion into a validated pre-experimental protocol, and the empirical demonstration that the protocol discriminates rather than uniformly flagging noise: at an identical budget ($N = 200$) it classifies one configuration as resolvable (observed range 10.5 pp against a threshold of 8.11 pp) and another as not.

Generality and honesty about limits. Validation spans fine-tuning of two model families across multiple seeds, scales and training paradigms; classical supervised learning (random forests on MNIST); and controlled synthetic simulation. An exploratory survey of 36 recent papers indicates that 71% of reported improvements fall below the resolution implied by their own evaluation budgets. The paper also reports a case from its own experiments in

which the diagnostic fails silently—a near-floor accuracy regime where the normal approximation breaks down and the threshold manufactures apparent resolvability—together with the operational guard that prevents it. A diagnostic that never reports its own inapplicability is of limited use to a practitioner, and I would rather delimit the framework than oversell it.

Fit with JDSA. The work sits within the journal’s remit of statistical and mathematical foundations for data science and analytics, and matches its transdisciplinary orientation: the diagnostic is model- and metric-agnostic, and I deliberately demonstrate it outside deep learning to establish that the limitation is a property of finite evaluation rather than an artifact of any one model class. It is also written to be usable—a lookup table and an open-source implementation let a reader run the check before committing an evaluation budget.

Declarations. The manuscript is original, has not been published previously, and is not under consideration elsewhere. I am the sole author. I declare no competing interests and received no funding for this work. All code and data supporting the findings, including the full literature-survey extraction table, are openly available at <https://github.com/wyp629929/A-Resolution-Framework-for-Finite-Sample-Model-Selection>.

Thank you for your time and consideration.

Sincerely,

Yaoping Wang